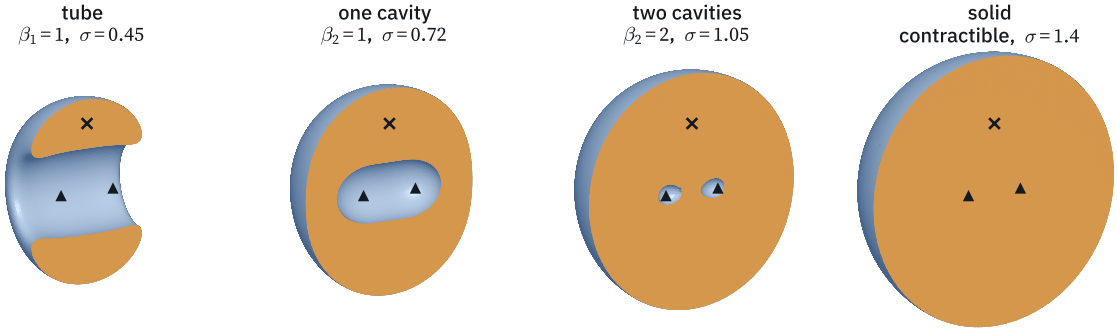

tube
$\beta_1 = 1,\ \sigma = 0.45$
one cavity
$\beta_2 = 1,\ \sigma = 0.72$
two cavities
$\beta_2 = 2,\ \sigma = 1.05$
solid
contractible, $\sigma = 1.4$